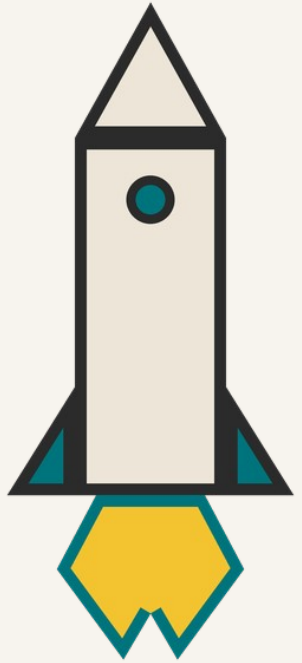
Daily Review



1. **State** a geostationary satellite's period.
2. **State** its altitude above Earth.
3. Altitude goes up: what happens to the period?
4. **Find** the weight of 40 kg on Earth.
5. **Give** two benefits of satellites.

Forces on Rockets

What it takes to leave the launch pad — and why it works.



National 5 Physics (S3)

What does it take to lift a rocket this big off the pad — and make it speed up?



Artemis II, 2026

PART 1

Newton's second law at lift-off

What makes a rocket speed up as it leaves the pad — and by how much.

By the end of this part...



1. I can **state Newton's second law** of motion
2. I can **apply** Newton's second law to **rockets**

Show me — describe each word



1. force
2. thrust
3. weight

Show me — describe each word



4 air resistance

5 unbalanced force

Show me — describe each word



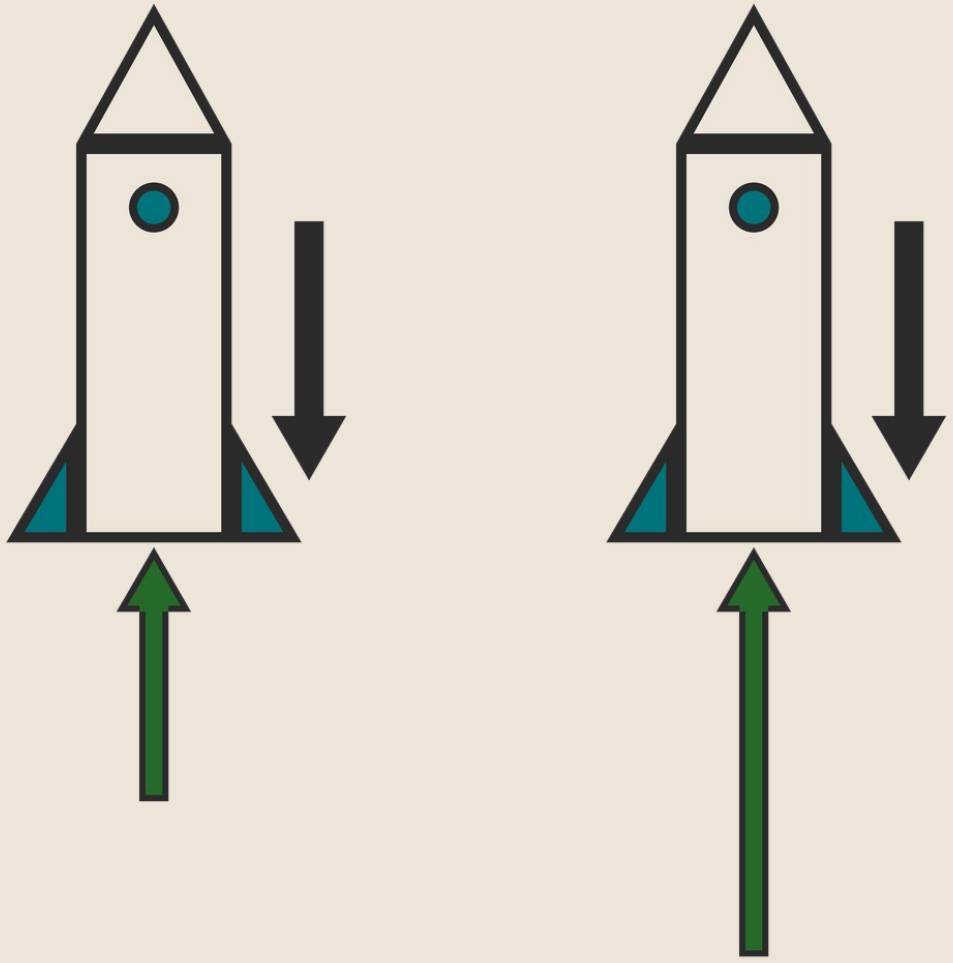
1. acceleration
2. Newton's first law
3. Newton's second law

Balanced, then unbalanced (p.15)

Balanced forces: a constant velocity, or staying still.

Thrust bigger than **weight**: an unbalanced force upwards.

An **unbalanced force** makes the rocket **accelerate**.

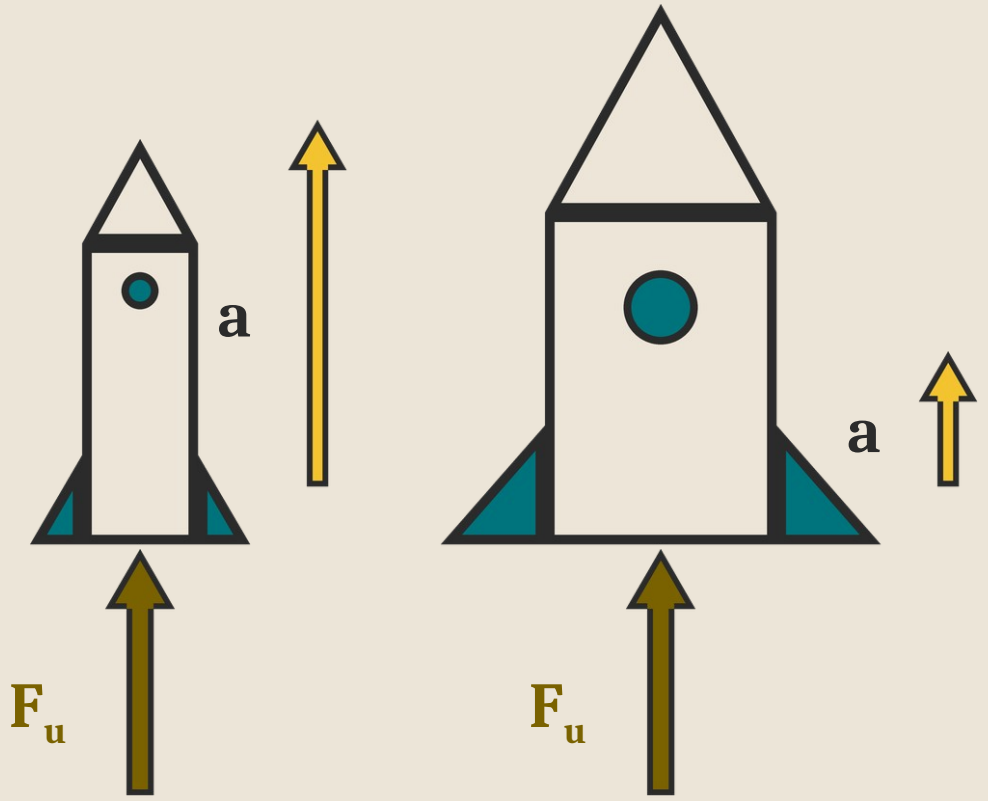


Force, mass and acceleration (p.15)

The same **unbalanced force** acts on both rockets.

The rocket with **less mass** has the **bigger acceleration**.

Burning fuel cuts the **mass**: same thrust, so **acceleration grows**.



Show me — Newton's laws

1. The forces on a rocket are **balanced**. What can it be doing?
2. What does an **unbalanced force** do to an object?
3. A rocket's **thrust** is **bigger** than its weight. What happens?

Show me — Newton's laws

4 Thrust **equals** weight. Is the rocket accelerating?

5 Two rockets have the same F_u. Which has the **bigger** a?

Task 1 — the first two laws (p.15)



Booklet page 15

Fill in the blanks for Newton's first and second laws.

Answers: Task 1 (p.15)

ANSWERS

Law	The missing words, in order
First law	forces … velocity
Second law	unbalanced

Unbalanced force, mass and acceleration (p.15)

$$F_u = m\,a$$

F_u	m	a
unbalanced force, in newtons (N)	mass, in kilograms (kg)	acceleration, in $m\,s^{-2}$

Task 2 — label the equation (p.15)

Booklet page 15

Label each letter in $F_u = ma$
with its quantity and its unit.

Answers: Task 2 (p.15)

ANSWERS

Letter	Quantity	Unit
F_u	unbalanced force	newton (N)
m	mass	kilogram (kg)
a	acceleration	$m\ s^{-2}$

Task 3 — forces at take-off (p.15)

Booklet page 15

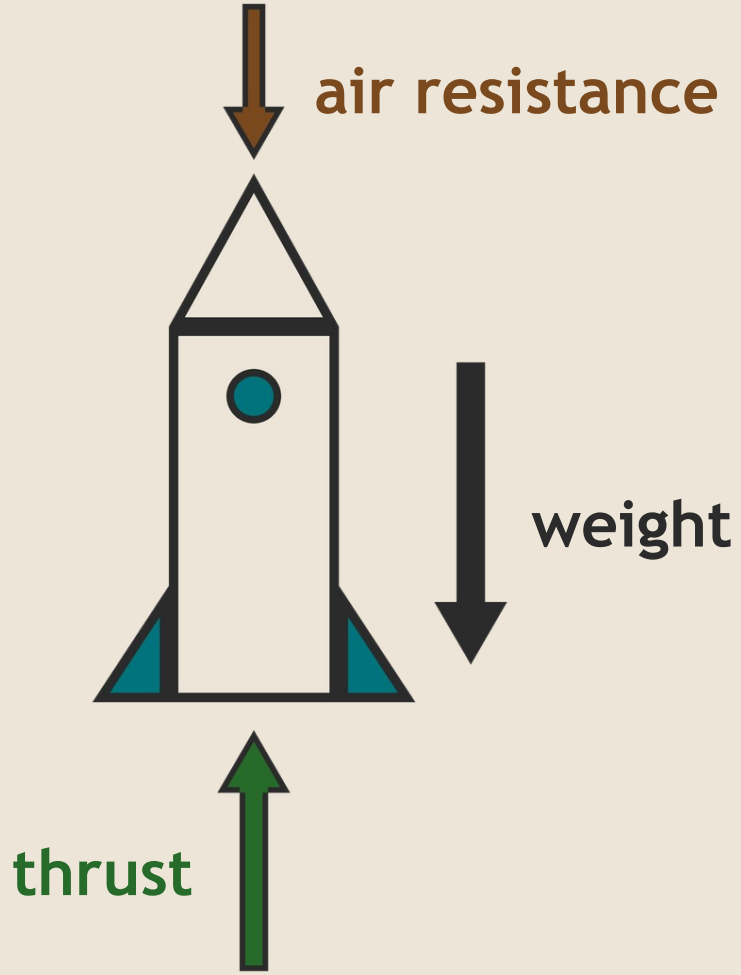
Label all the forces on the rocket at take-off.
Word bank: thrust, weight, air resistance, unbalanced force.

Answers: Task 3 (p.15)

Thrust acts upwards.

Weight and **air resistance** act downwards.

Unbalanced force = thrust − weight − air resistance.



Command words — 1 of 2

Word	What it asks for	A Space example
State	Name it, or give it in brief form.	State Newton's third law.
Describe	Give its features: what happens.	Describe the motion of a rocket with balanced forces.
Explain	Link cause and effect.	Explain how a rocket lifts off.

Command words — 2 of 2

Word	What it asks for	A Space example
Suggest	Use your physics in a new situation.	Suggest why a lander fires its engines before landing.
Predict	Say what may happen, from the facts given.	Predict how a changes as a rocket burns fuel.
Justify	Give reasons that back up your answer.	Is the rocket accelerating? Justify your answer.

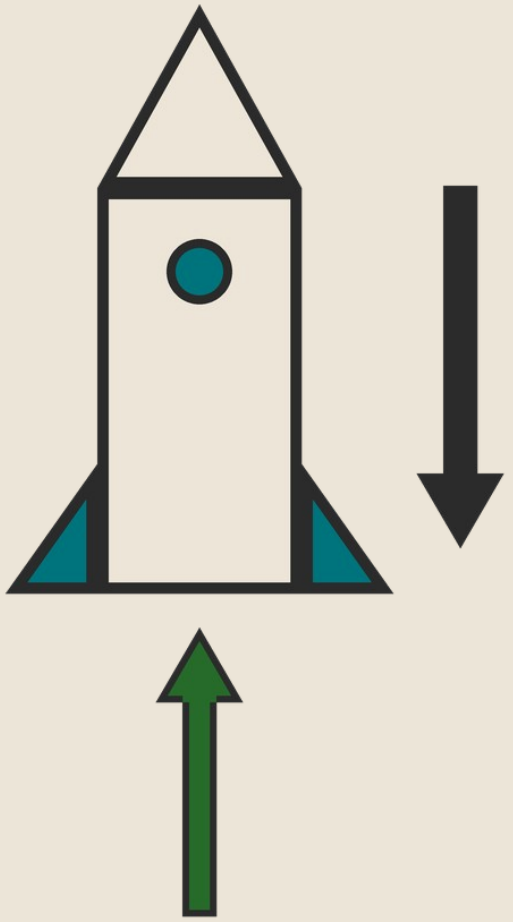
Worked example 1 — lift-off

1. A rocket of mass 2400 kg has a thrust of 36 000 N at lift-off. Air resistance can be ignored.

   (a) **Calculate** the **weight** of the rocket. **3**

   (b) **Determine** the **initial acceleration** of the rocket. **4**

No F_u worked out = only 1 mark out of 4.



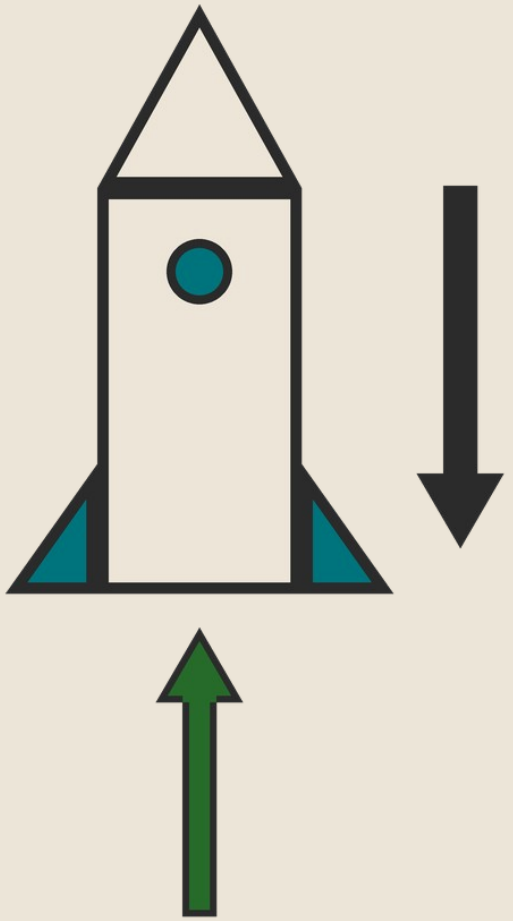
Your turn 1 — lift-off

2. A rocket of mass 1500 kg has a thrust of 27 000 N at lift-off. Air resistance can be ignored.

(a) **Calculate** the **weight** of the rocket. **3**

(b) **Determine** the **initial acceleration** of the rocket. **4**

No F_u worked out = only 1 mark out of 4.



Task 4 — try the example question (p.16)

Booklet page 16

A rocket has a mass of 1850 kg and its engine produces a thrust of 30 kN.
Draw the forces, then **calculate** the initial acceleration.

Answers: Task 4 (p.16)

ANSWERS

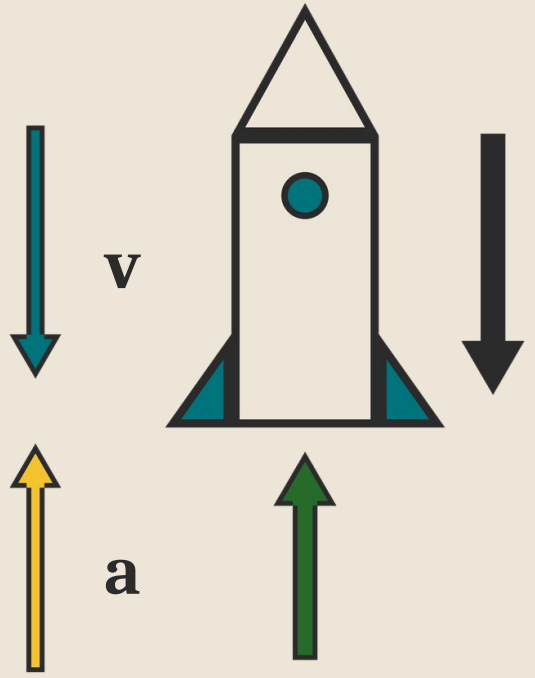
Step	Working
Thrust	30 kN = 30 000 N
Weight	W = 1850 × 9.8 = 18 130 N
Unbalanced force	F_u = 30 000 − 18 130 = 11 870 N
Acceleration	a = 11 870 / 1850 = 6.4 m s^{-2}

Worked example 2 — landing on Mars

3. A 400 kg space vehicle is falling towards Mars. Its engines apply an upward force of 2000 N. On Mars, $g = 3.7\ N\ kg^{-1}$.

(a) **Calculate** the **weight** of the vehicle on Mars. **3**

(b) **Determine** the **acceleration** of the vehicle. **4**



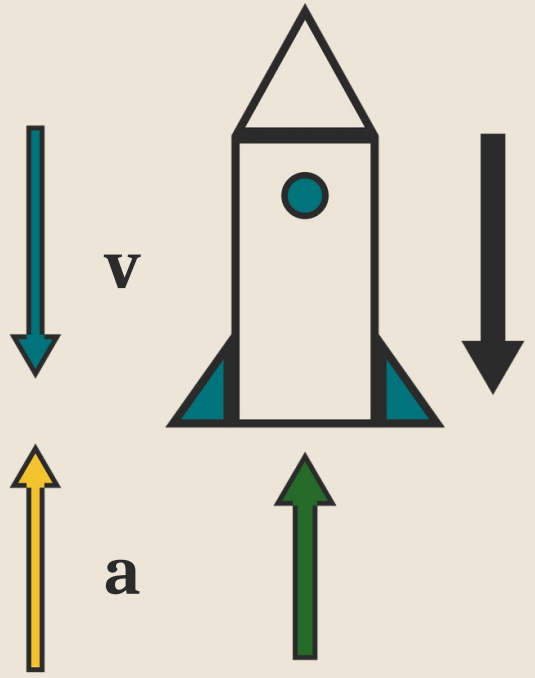
Moving down, a up: it slows down.

Your turn 2 — landing on the Moon

4. A 600 kg lander is falling towards the Moon. Its engines apply an upward force of 1500 N. On the Moon, g = 1.6 N kg^{-1}.

(a) **Calculate** the **weight** of the lander. **3**

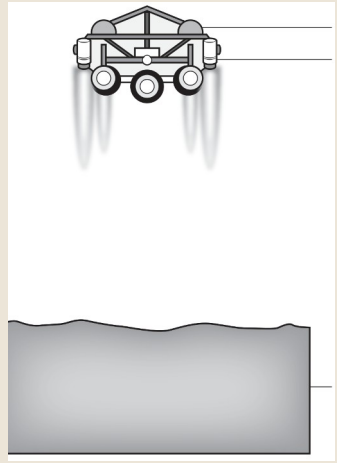
(b) **Determine** the **acceleration** of the lander. **4**



Moving down, a up: it slows down.

Hinge question — 2023 Section 1 Q8

8. A space vehicle of mass 350 kg is free falling vertically towards the surface of Mars.



Rocket engines are now fired, which apply a combined upwards force of 2200 N on the vehicle.

Just after the rocket engines are fired, the vehicle will

A move away from the surface of Mars at a constant speed

B move away from the surface of Mars with an increasing speed

C move towards the surface of Mars at a constant speed

D move towards the surface of Mars with a decreasing speed

E move towards the surface of Mars with an increasing speed.

2023

2025 Section 2 Q2(a)

2. The Falcon Heavy rocket is used to carry a satellite from Earth into space.

(a) (i) The total mass of the rocket and satellite at launch is 1.43×10^6 kg.
Calculate the **weight** of the rocket and satellite at launch. **3**

(ii) At launch, the initial upward thrust acting on the rocket is 2.28×10^7 N.
Determine the **initial acceleration** of the rocket and satellite. **4**


Hasbul Aerial Stock/Shutterstock

2025 paper

2025 Q2(a)(i) — marking instructions

Question	Expected response	Max mark	Additional guidance
2. (a)(i)	$W = mg$ (1 mark) $W = 1.43 \times 10^6 \times 9.8$ (1 mark) $W = 1.4 \times 10^7\ N$ (1 mark)	3	Accept: 1×10^7, 1.40×10^7, 1.401×10^7

2025 Q2(a)(ii) — marking instructions

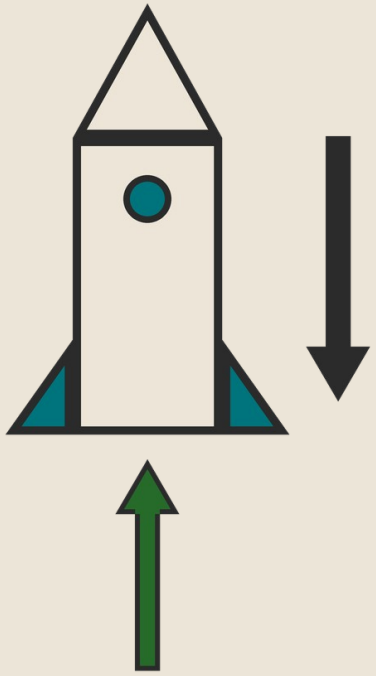
Question	Expected response	Max mark	Additional guidance
(a)(ii)	$F = 2.28 \times 10^7 - 1.4 \times 10^7$ (1 mark) $F = ma$ (1 mark) $2.28 \times 10^7 - 1.4 \times 10^7 = 1.43 \times 10^6 \times a$ (1 mark) $a = 6.2\ m\ s^{-2}$ (1 mark)	4	Or consistent with (a)(i). If no attempt to calculate the unbalanced force, then MAX (1) for the relationship. If clear arithmetic error in calculation of unbalanced force, then MAX (3). Accept: 6, 6.15, 6.154

2024 Section 2 Q4(c)

4. NASA is planning a crewed mission to the Moon.

(c) After spending time on the Moon, the astronauts will return to the orbiting spacecraft using the transportation module. During the first part of this return journey, the rockets on the transportation module will exert a constant upward force.
State what will happen to the **acceleration** of the transportation module during this part of the journey.
You must justify your answer.

3



2024 paper

2024 Q4(c) — marking instructions

Question	Expected response	Max mark	Additional guidance
4. (c)	Acceleration increases (1 mark) Justification: mass decreases (as fuel is used) OR gravitational field strength decreases (with altitude) (1 mark) weight decreases OR unbalanced force increases (1 mark)	3	MUST JUSTIFY Ignore any reference to air resistance/friction.

What Candidates Get Wrong

Thrust straight into F = ma

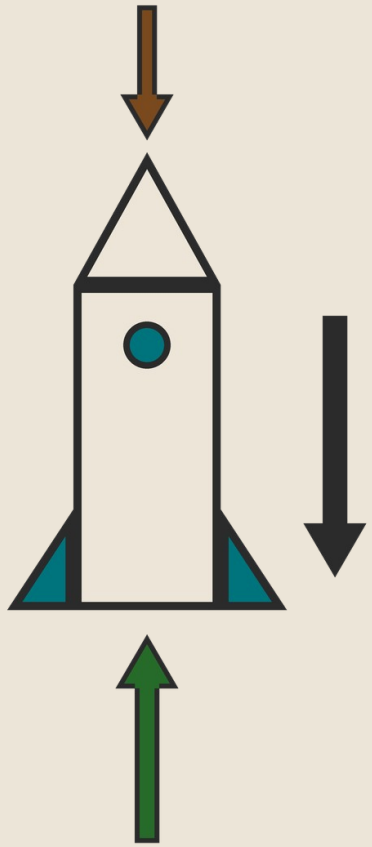
Find F_u = thrust − weight first. (2025 Q2(a)(ii) MI: no F_u, MAX 1 of 4)

F = ma for a weight

Weight is $W = mg$. (2024 course report)

A bare answer

Say why the acceleration changes. (2024 Q4(c) MI: MUST JUSTIFY)



Look back — by the end of this part...



1. I can **state Newton's second law** of motion
2. I can **apply** Newton's second law to **rockets**

Daily Review



1. What does an **unbalanced force** do?
2. **Calculate** a: 15 N on a 5.0 kg mass.
3. **Name** a rocket's upward force.
4. **Find** the weight of 70 kg on the Moon.
5. **State** Newton's second law.

PART 2

Newton pairs

Every force has a partner — and that is how a rocket flies.

By the end of this part...



1. I can **state Newton's third law** of motion
2. I can **apply** Newton's third law to **rockets**

Show me — describe each word



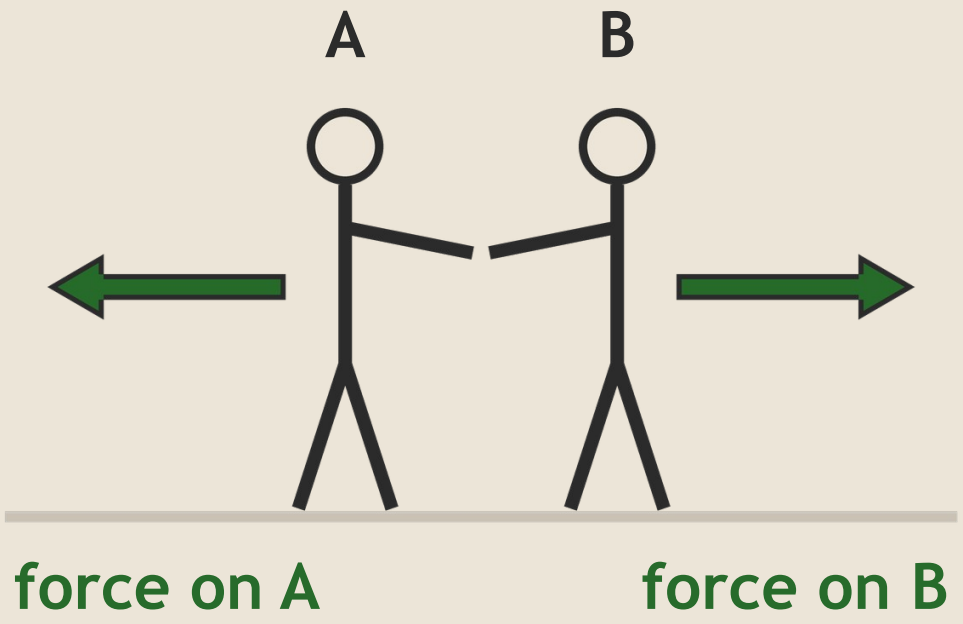
1. Newton's third law
2. Newton pair

Newton's third law (p.15)

If A exerts a force on B, B exerts an **equal** but **opposite** force on A.

Each force acts on a **different object**.

So the two forces **never cancel**.



Task 1 — the third law (p.15)



Booklet page 15

Fill in the blanks for Newton's third law.

Answers: Task 1 (p.15)

ANSWERS

Law	The missing words, in order
Third law	force … B … opposite … A

Agree, disagree or it depends?

1. Gases push on the **ground**, and that lifts the rocket.
2. The force on the rocket and the force on the gases **cancel out**.
3. The gases push on the rocket **as hard as** it pushes on them.

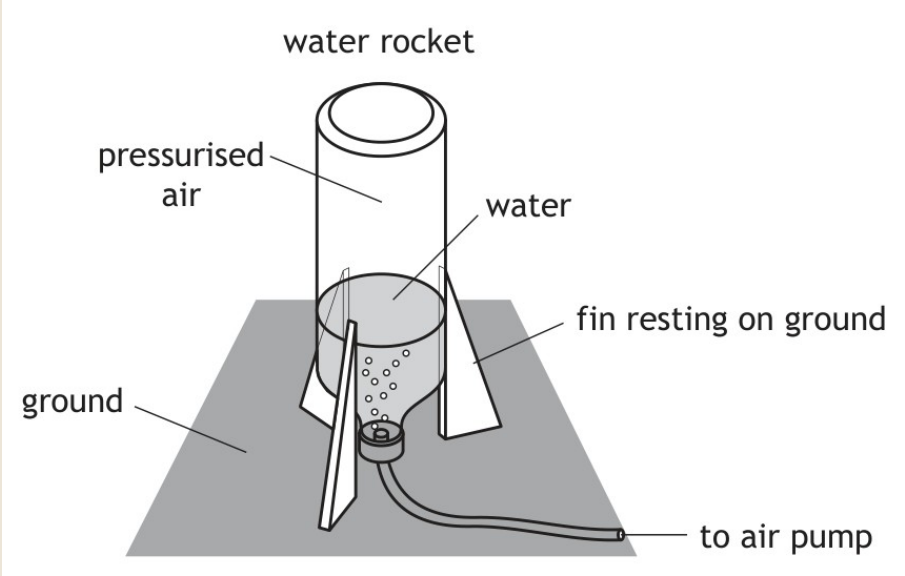
Agree, disagree or it depends?



4 The rocket pushes **harder** on the gases than they push back.

5 Thrust beats weight, so it **speeds up**.

Show me — the water rocket



What does the bottle push **downwards**?

What pushes the bottle **upwards**?

How do the two forces **compare**?

What must the thrust be **bigger** than?

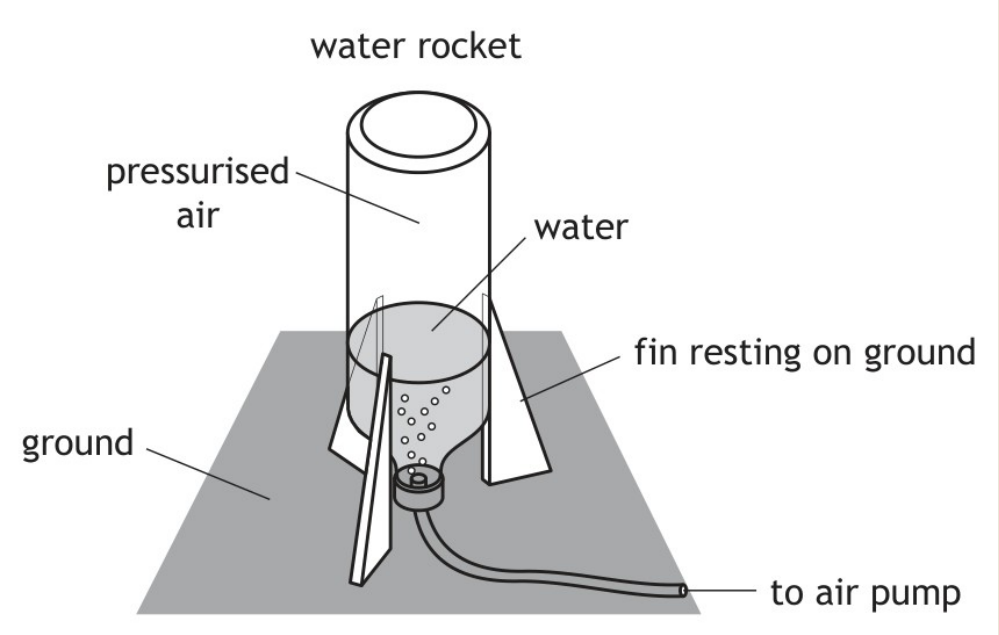
Demo — the water rocket

Predict: which way will the water go?

Predict: which way will the bottle go?

Watch the launch.

Explain it as a Newton pair: what pushes on what?



Task 5 — the Newton pair (p.16)

Booklet page 16

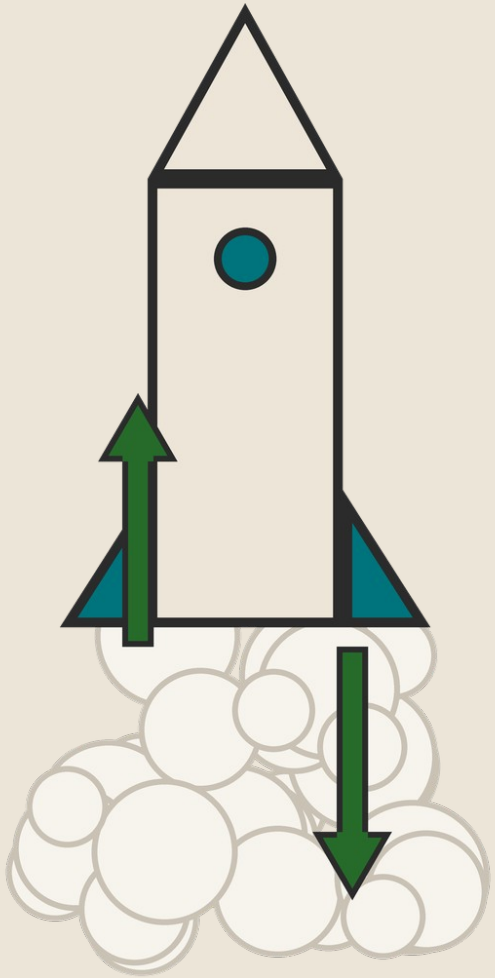
Complete the diagram to show the Newton pair acting on the rocket.

Answers: Task 5 (p.16)

Up: force on the rocket, due to the gases.

Down: force on the gases, due to the rocket.

Equal in size, **opposite** in direction.



A real engine — the RS-25



The engine pushes **hot gases** out.

The gases push back on the engine: **equal** and **opposite**.

No ground or air is needed, so it **works in space**.

Show me — Newton pairs

1. You push on a wall. What pushes **back** on you?

2. An astronaut throws a tool **forwards**. What happens to them?

3. A rocket pushes 8000 N on its gases. **State** the force on the rocket.

Show me — Newton pairs

4 Why do the two forces in a Newton pair **not cancel**?

5 **State** Newton's third law.

A rocket is taking off from the surface of the Earth. The rocket engines exert a force on the exhaust gases. Which of the following is the reaction to this force?



A The force of the Earth on the exhaust gases.

B The force of the Earth on the rocket engines.

C The force of the rocket engines on the Earth.

D The force of the exhaust gases on the Earth.

E The force of the exhaust gases on the rocket engines.

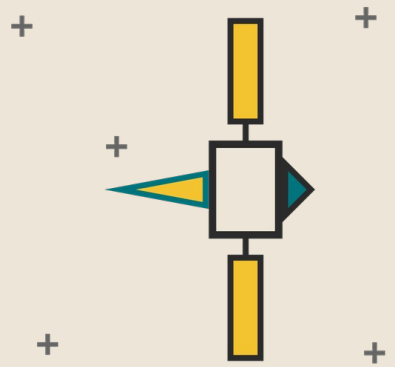
2024 Q3

2025 Section 2 Q4(b)(i)

4. In 2023, a spacecraft was launched from Earth to investigate the asteroid Psyche. Psyche is located in the asteroid belt, between Mars and Jupiter.

   (b) On its journey, the spacecraft uses an ion drive engine to provide thrust. The ion drive engine produces a small unbalanced force on the spacecraft.

   (i) **Explain** how this **small unbalanced force** can still result in a **large increase in speed**, even though the spacecraft has a large mass. **1**



2025 paper

2025 Q4(b)(i) — marking instructions

Question	Expected response	Max mark	Additional guidance
4. (b)(i)	(The unbalanced force) is applied over an extended/long period of time. (1 mark)	1	

Checkpoint



1. **State** Newton's second law.
2. Use Newton's third law to **explain** how a rocket lifts off.
3. Then the real exam question on the next slide: 2026 Section 2 Q4(b).

Checkpoint — 2026 Section 2 Q4(b)

4. A satellite is transported into space using a rocket.

(b) The total mass of the rocket and satellite at launch is 5.5×10^5 kg.

(i) **Show** that the **weight** of the rocket and satellite at launch is 5.4×10^6 N. **2**

(ii) The initial acceleration of the rocket and satellite at launch is 4.1 m s^{-2}.

Determine the initial upward **thrust** acting on the rocket. **4**



2026 paper

2026 Q4(b)(i) — marking instructions

Question	Expected response	Max mark	Additional guidance
4. (b)(i)	$W = mg$ (1 mark) $W = 5.5 \times 10^5 \times 9.8$ (1 mark) $W = 5.4 \times 10^6$ N	2	'Show' question Final answer of 5.4×10^6 N or its numerical equivalent, including unit, must be shown or MAX (1).

2026 Q4(b)(ii) — marking instructions

Question	Expected response	Max mark	Additional guidance
(b)(ii)	$F = ma$ (1 mark) $F = 5.5 \times 10^5 \times 4.1$ (1 mark) $(Th - W = F)$ $Th - 5.4 \times 10^6 = (5.5 \times 10^5 \times 4.1)$ (1 mark) $Th = 7.7 \times 10^6$ N (1 mark)	4	Accept: 8×10^6, 7.66×10^6, 7.655×10^6

What Candidates Get Wrong

Cost, not physics

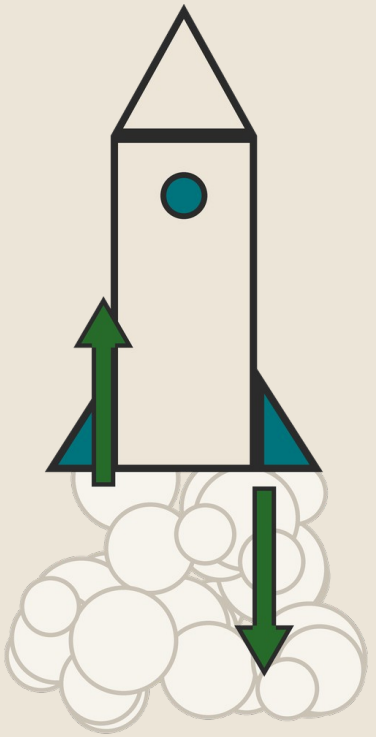
Use physics: lower g on the Moon, less energy to launch. (2023 Q4 report)

Small force, big speed

A small force for a long time: a big speed increase. (2025 Q4(b)(i) report)

"Gravity" or "upthrust"

Say weight or air resistance. (2025 Q2(b)(i) MI)

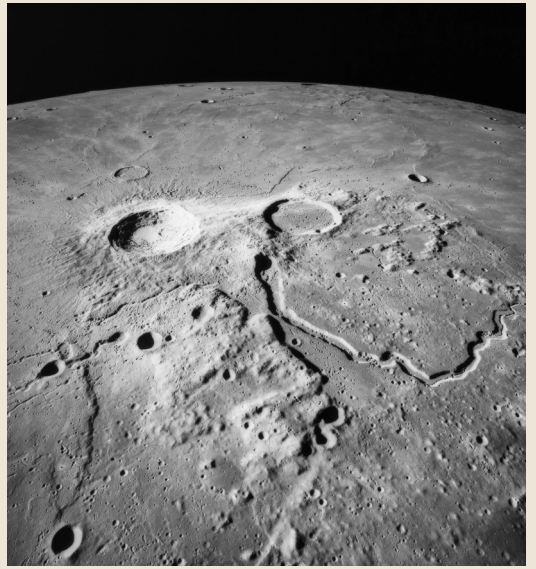


2023 Section 2 Q4 — open-ended

4. A space scientist makes the following statement.

‘Before we can have human space exploration of the solar system and beyond, we need to build a base on the Moon.’

Using your knowledge of physics, **comment on** the **benefits** and/or **challenges** of using a base on the Moon from which humans could explore the solar system and beyond. **3**

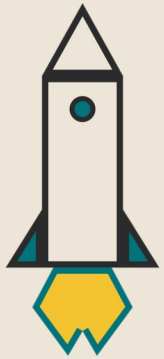


2023 paper

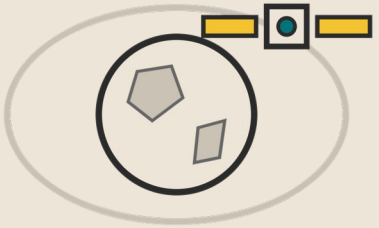
How the 3 marks are given

Marks	What the answer shows
3	Good understanding: the principles, a relationship, applied. It need not be complete.
2	Reasonable understanding: relevant statements.
1	Limited understanding: a little of the physics.
0	Nothing relevant, or repeats the question.

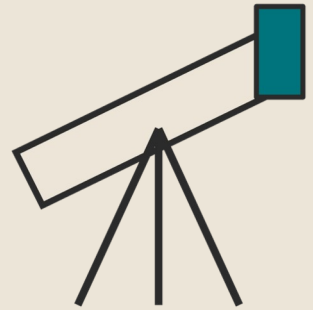
Careers — where this physics goes



Rocket engineer



Satellite engineer



Astronomer

UKube-1, Scotland's first satellite, was built in Glasgow.

SaxaVord Spaceport on Unst, Shetland, is licensed for launches.

Exam practice

Past papers

Year	Question
Specimen	Section 1 Q4
Specimen	Section 2 Q2(a), (c)
2025	Section 2 Q2(b)
Booklet	Page 17, Task 6

Specimen

Review 1 (p.18)



Booklet page 18

Check your booklet is up to date.
Test yourself on the definitions.
Do the Space practice questions on the website,
Parts A and B, and mark them.
Fill in your learning log.

Plenary

1. The big question: what does it take to lift a rocket off the pad?
2. **State** Newton's second law.
3. **State** Newton's third law.

Plenary

4 **Name** the forces on a rocket at take-off.

5 A 200 kg probe has an unbalanced force of 50 N. **Calculate** a.

Extension

1. A rocket's thrust stays the same as it burns fuel. **Explain** what happens to a.

2. **Explain** why a rocket works in space, where there is no air to push on.

3. A 20 000 kg rocket lifts off Earth at $1.2\ \mathrm{m\ s^{-2}}$. **Find** the thrust.

Look back — by the end of this part...



1. I can **state Newton's third law** of motion
2. I can **apply** Newton's third law to **rockets**

LOOKING AHEAD

The Big Bang Theory

How did the universe begin — and how do we know how old it is?

Course specification — 1 of 3

- Knowledge of Newton’s second and third laws and their application to space travel, rocket launch and landing.

Course specification, page 8

Course specification — 2 of 3

- Application of Newton’s laws and unbalanced forces to explain and/or determine acceleration for situations where more than one force is acting.
- Use of an appropriate relationship to solve problems involving unbalanced force, mass and acceleration, for situations where one or more forces are acting in one dimension or at right angles.

Course specification — 3 of 3

- $F = ma$
- Explanation of motion resulting from a ‘reaction’ force in terms of Newton’s third law.

Credits (1 of 3)

The big question (Artemis II, 2026): NASA/Michael DeMocker

Command words — 1 of 2: © Qualifications Scotland (SQA). N5 Physics specimen marking instructions, p.03

Command words — 2 of 2: © Qualifications Scotland (SQA). N5 Physics specimen marking instructions, p.03

Hinge question — 2023 Section 1 Q8: © Qualifications Scotland (SQA). 2023 N5 Physics Section 1, Q8

2025 Section 2 Q2(a): © Qualifications Scotland (SQA). 2025 N5 Physics Section 2, Q2(a); marking instructions. Photo in the paper: Hasbul Aerial Stock/Shutterstock

2025 Q2(a)(ii) — marking instructions: © Qualifications Scotland (SQA). 2025 N5 Physics Section 2, Q2(a); marking instructions. Photo in the paper: Hasbul Aerial Stock/Shutterstock

2024 Section 2 Q4(c): © Qualifications Scotland (SQA). 2024 N5 Physics Section 2, Q4(c); marking instructions

Credits (2 of 3)

What Candidates Get Wrong: © Qualifications Scotland (SQA). 2025 N5 Physics course report; 2025 marking instructions; © Qualifications Scotland (SQA). 2024 N5 Physics course report; 2024 marking instructions; © Qualifications Scotland (SQA). 2023 N5 Physics course report

Show me — the water rocket: © Qualifications Scotland (SQA). N5 Physics 2014, Section 2 Q12.

Demo — the water rocket: © Qualifications Scotland (SQA). N5 Physics 2014, Section 2 Q12.

A real engine — the RS-25: NASA

Hinge question — 2024 Section 1 Q3: © Qualifications Scotland (SQA). 2024 N5 Physics Section 1, Q3

2025 Section 2 Q4(b)(i): © Qualifications Scotland (SQA). 2025 N5 Physics Section 2, Q4(b)(i); marking instructions

Checkpoint — 2026 Section 2 Q4(b): © Qualifications Scotland (SQA). 2026 N5 Physics Section 2, Q4(b); marking instructions p.13. Image in the paper: Nerthuz/shutterstock.com

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R Stewart - November 2026